

The empirical recurrence for OEIS A235196, proved by transfer matrix

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Abstract

OEIS A235196 carries an empirical recurrence of order 39 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is local to every 2×2 subblock, hence two consecutive lines, so the admissible configurations of such a window are the vertices of a finite digraph and the arrays counted are exactly the walks in it; hence $a(n)$ is a walk count on $S = 326592$ vertices and is C -finite. Whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A235196 is “Number of $(n+1) \times (6+1)$ 0..6 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 5, with no adjacent elements equal (constant-stress tilted 1×1 tilings)”. It has offset 1 and begins

98080, 176104, 333340, 774140, 1816160, 4812096, 12604104, 35505020, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 9a(n-1) + 57a(n-2) - 731a(n-3) - 878a(n-4) + 26304a(n-5) - 15900a(n-6) - 552844a(n-7) + 899723a(n-8) + 7521105a(n-9) - 18125633a(n-10) - 69122561a(n-11) + 220983779a(n-12) + 431291855a(n-13) - 1831340243a(n-14) - 1746606119a(n-15) + 10803108780a(n-16) + 3716404250a(n-17) - 46315168788a(n-18) + 2550194940a(n-19) + 145490417695a(n-20) - 46954531755a(n-21) - 335197745077a(n-22) + 166429786799a(n-23) + 564880422518a(n-24) - 334400168840a(n-25) - 694120308016a(n-26) + 424547635156a(n-27) + 620744392928a(n-28) - 343613520080a(n-29) - 401608244880a(n-30) + 170947437904a(n-31) + 182786646240a(n-32) - 46865973504a(n-33) - 54609655680a(n-34) + 4665398976a(n-35) + 9369063936a(n-36) + 551978496a(n-37) - 679311360a(n-38) - 113218560a(n-39).$

As of the “Last modified” line on the live entry (Jun 19 2022 03:33:22, revision 10) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The condition the entry imposes is local: it constrains every 2×2 subblock, hence two consecutive lines and nothing beyond. Take as vertices the admissible configurations of one such window and put an edge from u to v whenever v may follow u , which the condition decides by inspection

